PRAJNA

Exam Intelligence Platform

Technical Architecture Document

v1.0 · March 2026

NEET · JEE Main · JEE Advanced

23,119 Questions · 2000–2025

23 M-param SLM · FastAPI · RAG

PLATFORM AT A GLANCE

Key metrics and scope

23,119
Questions in database

25 yrs
Coverage: 2000–2025

3 exams
NEET · JEE Main · JEE Advanced

23 M
SLM Parameters (PRAJNA)

399-dim
Input feature vector

12
REST API endpoints

≥60%
Top-10 coverage

≥85%
Grounding accuracy

≤3 s
API latency P95

≤10%
Fallback rate

THREE-TIER ARCHITECTURE

TIER 1

Data & ML Engine

- SQLite exam.db (10.9 MB)
- 23,119 questions
- PRAJNA SLM (.pt models)
- Predictor v3 (3-stage)
- Student CSVs (37+ MB)

Python · PyTorch · Pandas

TIER 2

Intelligence Layer

- FastAPI :8001
- ChromaDB :8002 (RAG)
- Ollama :11434 (SLM)
- Pydantic schemas
- Anti-hallucination guard

FastAPI · ChromaDB · Sentence-Transformers

TIER 3

Frontend

- Streamlit :8501
- Static HTML dashboard
- Chart.js · Client-side JS
- Pre-built JSON summaries
- Zero-server mode

Streamlit · Vanilla JS · Chart.js

DATA LAYER

SQLite SCHEMA

Column	Type	Notes
id	TEXT PK	"JEE_ADV_2019_P1_Q12"
exam	TEXT	NEET | JEE Main | JEE Adv
year	INTEGER	2000 – 2025
subject	TEXT	Physics / Chemistry / Bio / Math
topic	TEXT	Chapter-level topic name
micro_topic	TEXT	Specific sub-topic
question_type	TEXT	MCQ / integer / matrix_match
difficulty	INTEGER	1 – 5 scale

DATA FILES

exam.db	10.9 MB
neet_results_v2.csv	16 MB
jee_results_v2.csv	21.4 MB
students_v2.csv	14 KB
student_summary_*.json	1.7 MB

INGESTION PIPELINE

PDF/HTML exam papers → **Claude extraction prompt** → JSON
JSON →
loader.py → **exam.db** (deduplication on ingest)

PRAJNA SLM — MODEL ARCHITECTURE

Chapter Name
(text input)

→

all-MiniLM-L6-v2
384-dim embedding

→

15 Temporal
Features

+

Concat → 399d
Input Vector

MLP LAYERS

Dense 256
ReLU + BN + Drop 0.2

Dense 128
ReLU + BN + Drop 0.2

Dense 64
ReLU

23 M Parameters

OUTPUT HEADS

Head 1 — P(appear)

Sigmoid → probability
chapter will appear

Head 2 — E[questions]

Linear regression
expected Q count

Head 3 — Difficulty

Softmax → 5-class
(1 = easy … 5 = hard)

FEATURE ENGINEERING — 15 TEMPORAL FEATURES

0	**Appearance rate**	count(years appeared) / total span
1	**Recency-weighted freq**	Σ 1/(year_diff^1.5 + 1) — exp decay
2	**Recent presence (3yr)**	Binary: appeared in last 3 years?
3	**Recent presence (5yr)**	Binary: appeared in last 5 years?
4	**Gap (years since last)**	Current year – last appearance year
5	**Mean gap**	Avg interval between appearances
6	**Trend slope**	Linear regression coeff on last 10 years
7	**Mean question count**	Avg questions per appearing year
8	**Std-dev question count**	Variance in annual question counts
9	**Max question count**	Historical peak in a single year
10	**Total appearances**	Normalised by total year window
11	**Cross-exam presence**	1.0 if ≥1 other exam type covers it
12	**Cycle regularity**	Inverse CV of appearance gaps
13	**Recent yield avg**	Mean questions in last 3 appearances
14	**Subject encoding**	0.25=Phys 0.50=Chem 0.75=Bio 1.0=Math

PREDICTION PIPELINE — PREDICTOR v3

01 Stage 1 Appearance Probability

- Recency-weighted frequency
- Gap-return probability
- Trend slope (lin. regression)
- Cycle-match score

02 Stage 2 Weightage Model

- Expected question count
- Historical Q-count distribution
- Accounts for difficulty trends
- Given chapter appears

03 Stage 3 Question Format

- Likely question types
- MCQ single / MCQ multi
- Integer / Numerical
- Matrix match

POST-PROCESSING

- Subject balance quotas
- Unique chapter enforcement
- Diversification penalty
- Constraint-solving reranker

INTELLIGENCE LAYER — FastAPI SERVICE

REST ENDPOINTS

Method	Path	Purpose
GET	/predictions/rank-all	Global chapter ranking
GET	/predictions/chapter/{name}	Chapter-level breakdown
POST	/insights/micro-topic	SLM micro-topic explanation
POST	/insights/chapter	Chapter intelligence + clusters
POST	/reports/revision-plan	Personalised revision plan
POST	/copilot/ask	Natural language Q&A
GET	/data/trends	Trending topic briefing
GET	/health	Service health check

SLM PROVIDERS

mock	Zero-setup (fixtures)
ollama	Local mistral:7b
huggingface	phi-3-mini (local GPU)
openai_compat	GPT-4o-mini / Groq / vLLM

KEY SERVICES

TopicAggregator	Batch builder, priority scorer, ranker
ClusterDetector	Semantic topic cluster detection
RAGRetriever	Top-K chroma retrieval, metadata filter
AntiHallucGuard	Fact grounding + fallback trigger

RAG + ANTI-HALLUCINATION GUARD

User Query → Embed (384-dim) → ChromaDB Top-K Search → Metadata Filter → SLM Prompt + Context → Guard Validation → Response

RAG CONFIGURATION

Vector Store:	ChromaDB (local :8002 or managed)
Embedding:	all-MiniLM-L6-v2 — 384-dim, CPU
Top-K:	K = 5 documents per query (configurable)
Min Relevance:	0.35 cosine similarity threshold
Metadata Filter:	exam · year · subject · micro_topic
Context Injection:	Retrieved docs prepended to SLM prompt

ANTI-HALLUCINATION

≥ 0.85	Grounding accuracy target All numerical facts verified vs. DB
≥ 0.75	Factual consistency target Claims cross-checked multi-source
≤ 10%	Fallback rate target Template response if confidence fails

FRONTEND ARCHITECTURE

STREAMLIT DASHBOARD :8501

- Multi-page: Insights · Deep Dive · Copilot · API Docs
- Live SQLite integration + Plotly charts
- Student list with search + batch filter
- Hero card: accuracy %, strength zones, score
- Ranked prediction cards with urgency badges
- Copilot chat (streamed SLM responses)
- Dark theme — #0F0F1A / #7C3AED

STATIC HTML DASHBOARD (no server)

- Pure vanilla JS — zero framework dependency
- Loads pre-built JSON summaries (903 KB NEET)
- Level system: M ≥80% S ≥65% D ≥45% W ≥25% C <25%
- Priority score: (100−accuracy) × (slmImportance/100)
- Lazy guide: 59 chapter cards built on first expand
- Micro-topic API with Map-based dedup cache
- Chart.js radar + bar charts

DEPLOYMENT & INFRASTRUCTURE

Local Dev

- python run.py
- uvicorn :8001 --reload
- open student-dashboard.html

Docker Compose

- API :8001
- Ollama :11434
- ChromaDB :8002

Railway

- railway.toml config
- Streamlit-first
- Git-push deploy

ngrok Tunnel

- python serve.py
- Public URL generated
- Demo-ready sharing

Key env vars: SLM_PROVIDER · OLLAMA_MODEL · RAG_ENABLED · CHROMA_HOST · PREDICTION_CACHE_TTL_SEC · CORS_ORIGINS

QUALITY METRICS & ROADMAP

QUALITY METRICS

Coverage@10	60%
Heavy-Topic Recall	75%
Rank Correlation τ	≥0.70
Grounding Accuracy	85%
Student Usefulness	80%

ROADMAP

Phase 1	Foundation	✓ Done
Phase 2	Intelligence API	✓ Done
Phase 3	Student Personalisation	✓ Done
Phase 4	Production Hardening	→ In Progress
Phase 5	Advanced Features	◷ Planned

PRAJNA

Built to make every study hour count.

- 23,119 questions · 25 years · 3 exam types
- 23M-param SLM with 15 temporal features
- FastAPI + RAG + pluggable SLM providers
- Static + Streamlit dual frontend